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Applied Calculus IV

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Week 14

Chapter 7



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Definition (Laplace Transform)

Let f be a function defined for $t \geq 0$. The **Laplace transform** of f , denoted $\mathcal{L}\{f(x)\}$, is given by:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

when the integral **converges**.

Exercise: Show that if $\mathcal{L}\{f(t)\} = F(s)$ and a is any real number, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a).$$

Solution:

By definition,

$$\mathcal{L}\{e^{at}f(t)\} = \int_0^{\infty} e^{-st}e^{at}f(t) dt = \int_0^{\infty} e^{-(s-a)t}f(t) dt = F(s-a).$$

Example: Evaluate $\mathcal{L}\{e^{5t}t^3\}$

Solution:

$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} \xrightarrow{s \rightarrow s-5} \mathcal{L}\{e^{5t}t^3\} = \frac{3!}{(s-5)^4}$$

Example: Evaluate $\mathcal{L}\{e^{-2t} \cos 4t\}$

Solution:

$$\mathcal{L}\{\cos 4t\} = \frac{s}{s^2 + 16} \xrightarrow{s \rightarrow s+2} \mathcal{L}\{e^{-2t} \cos 4t\} = \frac{(s+2)}{(s+2)^2 + 16}$$

Example: Evaluate

$$\mathcal{L}^{-1} \left\{ \frac{7s + 4}{s^2 + 4s + 6} \right\}$$

Solution:

$$\mathcal{L}^{-1}\left\{\frac{7s+4}{s^2+4s+6}\right\} = \mathcal{L}^{-1}\left\{\frac{7s+14-10}{(s+2)^2+2}\right\} = 7\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+2}\right\} - 10\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+2}\right\}$$

$s+2 \rightarrow s$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+2}\right\} = \cos \sqrt{2}t$$

$s+2 \rightarrow s$

$$\mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2+2}\right\} = \sin \sqrt{2}t$$

➡
$$\mathcal{L}^{-1}\left\{\frac{s+5}{s^2+4s+6}\right\} = 7e^{-2t} \cos \sqrt{2}t - 5\sqrt{2}e^{-2t} \sin \sqrt{2}t$$

Example: Evaluate

$$\mathcal{L}^{-1} \left\{ \frac{2s + 5}{(s - 3)^2} \right\}$$

Solution:

$$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{2s-6+11}{(s-3)^2}\right\} = 2\mathcal{L}^{-1}\left\{\frac{\cancel{s-3}}{(s-3)^2}\right\} + 11\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\}$$

$s-3 \rightarrow s$ $s-3 \rightarrow s$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1 \qquad \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$$

➡ $\mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = 2e^{3t} + 11te^{3t}$

In Problem 1-10, evaluate $F(s)$

1. $\mathcal{L}\{te^{10t}\}$

2. $\mathcal{L}\{te^{-6t}\}$

3. $\mathcal{L}\{t^3e^{-2t}\}$

4. $\mathcal{L}\{t^{10}e^{-7t}\}$

5. $\mathcal{L}\{t(e^t + e^{2t})^2\}$

6. $\mathcal{L}\{e^{2t}(t - 1)^2\}$

7. $\mathcal{L}\{e^t \sin 3t\}$

8. $\mathcal{L}\{e^{-2t} \cos 4t\}$

9. $\mathcal{L}\{(1 - e^t + 3e^{-4t}) \cos 5t\}$

10. $\mathcal{L}\left\{e^{3t}\left(9 - 4t + 10 \sin \frac{t}{2}\right)\right\}$

In Problem 10-20, evaluate $f(t)$

$$11. \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^3}\right\}$$

$$12. \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\}$$

$$13. \mathcal{L}^{-1}\left\{\frac{1}{s^2 - 6s + 10}\right\}$$

$$14. \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 2s + 5}\right\}$$

$$15. \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4s + 5}\right\}$$

$$16. \mathcal{L}^{-1}\left\{\frac{2s + 5}{s^2 + 6s + 34}\right\}$$

$$17. \mathcal{L}^{-1}\left\{\frac{s}{(s+1)^2}\right\}$$

$$18. \mathcal{L}^{-1}\left\{\frac{5s}{(s-2)^2}\right\}$$

$$19. \mathcal{L}^{-1}\left\{\frac{2s-1}{s^2(s+1)^3}\right\}$$

$$20. \mathcal{L}^{-1}\left\{\frac{(s+1)^2}{(s+2)^4}\right\}$$

More Examples ...

Example: Solve the equation $x'' + 2x' + x = e^{-t}$, with $x(1) = x'(1) = 0$.



Recall

$$\mathcal{L}[x'] = s\mathcal{L}[x] - x(0)$$

$$\mathcal{L}[x''] = s^2\mathcal{L}[x] - sx(0) - x'(0)$$

Solution:

Let $\tau = t - 1$:

$$x''(\tau) + 2x'(\tau) + x(\tau) = e^{-(\tau+1)}, \quad x(0) = x'(0) = 0.$$

Take the Laplace transform:

$$s^2X(s) + 2sX(s) + X(s) = \frac{1}{s+1} \cdot \frac{1}{e} \quad \Rightarrow \quad X(s) = \boxed{\frac{1}{(s+1)^3}} \cdot \frac{1}{e}$$

$s+1 \rightarrow s$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2}\tau^2 \quad \longrightarrow \quad \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^3}\right\} = \frac{1}{2}e^{-t}\tau^2$$



$$x(\tau) = \frac{1}{2}\tau^2 e^{-\tau-1}$$

Solution:

And finally, $x(t)$ is

$$x(t) = \frac{1}{2}(t-1)^2 e^{-t}.$$

Example: Use the Laplace transform to solve the initial-value problem

$$x''' + 3x'' + 3x' + x = 1, \quad x(0) = 0, \quad x'(0) = 0, \quad x''(0) = 0$$

Recall

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0)$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0)$$

$$\mathcal{L}\{x'''(t)\} = s^3X(s) - s^2x(0) - sx'(0) - x''(0)$$

Solution:

Since $x(0) = x'(0) = x''(0) = 0$,

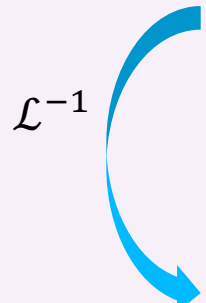
$$\begin{array}{l} \mathcal{L} \quad x''' + 3x'' + 3x' + x = 1 \\ \quad \underbrace{[s^3 X(s)]}_{x'''} + 3 \underbrace{[s^2 X(s)]}_{x''} + 3 \underbrace{[s X(s)]}_{x'} + \underbrace{X(s)}_x = \frac{1}{s} \end{array}$$

$$X(s) (s^3 + 3s^2 + 3s + 1) = \frac{1}{s}$$

$$\Rightarrow X(s) = \frac{1}{s(s+1)^3}$$

Solution:

Decompose the fraction $\frac{1}{s(s+1)^3} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{D}{(s+1)^3}$


$$X(s) = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} - \frac{1}{(s+1)^3}$$
$$x(t) = 1 - e^{-t} - te^{-t} - \frac{1}{2}t^2e^{-t}$$

Example: Use Laplace transforms to solve the given system of differential equations

$$\begin{cases} x_1' = 3x_1 + 5x_2 + e^{-t}, \\ x_2' = -5x_1 + 3x_2 \end{cases}$$

with the initial conditions

$$x_1(0) = 0, \quad x_2(0) = 1$$

Solution:

Let $X_1(s) = \mathcal{L}[x_1(t)]$ and $X_2(s) = \mathcal{L}[x_2(t)]$

$$\begin{cases} x_1' = 3x_1 + 5x_2 + e^{-t}, \\ x_2' = -5x_1 + 3x_2 \end{cases} \quad \Rightarrow \quad \begin{cases} sX_1(s) = 3X_1(s) + 5X_2(s) + \frac{1}{s+1}, \\ sX_2(s) - 1 = -5X_1(s) + 3X_2(s) \end{cases}$$

or

$$\begin{cases} (s-3)X_1(s) - 5X_2(s) = \frac{1}{s+1}, \\ 5X_1(s) + (s-3)X_2(s) = 1 \end{cases}$$

Cramer's Rule ($A\mathbf{X} = \mathbf{b}$)

$$A = \begin{pmatrix} s-3 & -5 \\ 5 & s-3 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ s+1 \end{pmatrix}$$

$$A_1 = \begin{pmatrix} 1 & -5 \\ s+1 & s-3 \end{pmatrix} \Rightarrow X_1 = \frac{\det(A_1)}{\det(A)} = \frac{\frac{s-3}{s+1} + 5}{(s-3)^2 + 5^2},$$

$$A_2 = \begin{pmatrix} s-3 & 1 \\ 5 & s+1 \end{pmatrix} \Rightarrow X_2 = \frac{\det(A_2)}{\det(A)} = \frac{s-3 - \frac{5}{s+1}}{(s-3)^2 + 5^2}$$

$$\Rightarrow \begin{cases} X_1(s) = \frac{\frac{s-3}{s+1} + 5}{(s-3)^2 + 5^2} \\ X_2(s) = \frac{s-3 - \frac{5}{s+1}}{(s-3)^2 + 5^2} \end{cases}$$

Decompose
the fractions

$$\begin{cases} X_1(s) = \frac{6s+2}{(s+1)[(s-3)^2+25]} \\ X_2(s) = \frac{(s-4)(s+2)}{(s+1)[(s-3)^2+25]} \end{cases}$$

$$\begin{cases} X_1(s) = \frac{1}{41} \left[4 \frac{s-3}{(s-3)^2+5^2} + 46 \frac{5}{(s-3)^2+5^2} - 4 \frac{1}{s+1} \right] \\ X_2(s) = \frac{1}{41} \left[46 \frac{s-3}{(s-3)^2+5^2} - 4 \frac{5}{(s-3)^2+5^2} - 5 \frac{1}{s+1} \right] \end{cases}$$

$\mathcal{L}^{-1}$

$$\begin{cases} x_1(t) = \frac{4}{41} e^{3t} \cos 5t + \frac{46}{41} e^{3t} \sin 5t - \frac{4}{41} e^{-t}, \\ x_2(t) = \frac{46}{41} e^{3t} \cos 5t - \frac{4}{41} e^{3t} \sin 5t - \frac{5}{41} e^{-t} \end{cases}$$

More details:

$$X_1(s) = \frac{1}{41} \left[4 \frac{s-3}{(s-3)^2 + 5^2} + 46 \frac{5}{(s-3)^2 + 5^2} - 4 \frac{1}{s+1} \right]$$

$$s-3 \rightarrow s$$

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 5^2} \right\} = \cos 5t$$



$$\mathcal{L}^{-1} \left\{ \frac{s-3}{(s-3)^2 + 5^2} \right\} = e^{3t} \cos 5t$$

$$s-3 \rightarrow s$$

$$\mathcal{L}^{-1} \left\{ \frac{5}{s^2 + 5^2} \right\} = \sin 5t$$



$$\mathcal{L}^{-1} \left\{ \frac{5}{(s-3)^2 + 5^2} \right\} = e^{3t} \sin 5t$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s - (-1)} \right\} = e^{-t}$$

$$\longrightarrow x_1(t) = \frac{4}{41} e^{3t} \cos 5t + \frac{46}{41} e^{3t} \sin 5t - \frac{4}{41} e^{-t}$$

And similarly for $X_2(s)$

Example: Use Laplace transforms to solve the given system of differential equations

$$\begin{cases} x_1' = 2x_1 + x_2 \\ x_2' = -x_1 + 4x_2 \end{cases}$$

with the initial conditions

$$x_1(0) = 0, x_2(0) = 1$$

Solution:

Let $X_1(s) = \mathcal{L}[x_1(t)]$ and $X_2(s) = \mathcal{L}[x_2(t)]$

$$\mathcal{L} \left\{ \begin{cases} x_1' = 2x_1 + x_2 \\ x_2' = -x_1 + 4x_2 \end{cases} \right.$$

$$\begin{cases} sX_1(s) - \cancel{x_1(0)} = 2X_1(s) + X_2(s) \\ sX_2(s) - \cancel{x_2(0)} = -X_1(s) + 4X_2(s) \end{cases}$$

1

$$\Rightarrow \begin{cases} (s-2)X_1(s) - X_2(s) = 0 \\ X_1(s) + (s-4)X_2(s) = 1 \end{cases}$$

Cramer's Rule ($A\mathbf{X} = \mathbf{b}$)

$$\det(A) = \begin{vmatrix} s-2 & -1 \\ 1 & s-4 \end{vmatrix} = (s-2)(s-4) + 1 = s^2 - 6s + 9 = (s-3)^2$$

$$X_1(s) = \frac{\begin{vmatrix} 0 & -1 \\ 1 & s-4 \end{vmatrix}}{\det(A)} = \frac{0 \cdot (s-4) - (-1) \cdot 1}{(s-3)^2} = \frac{1}{(s-3)^2}$$

$$X_2(s) = \frac{\begin{vmatrix} s-2 & 0 \\ 1 & 1 \end{vmatrix}}{\det(A)} = \frac{(s-2) \cdot 1 - 0 \cdot 1}{(s-3)^2} = \frac{s-2}{(s-3)^2} \quad \frac{(s-3)+1}{(s-3)^2} = \frac{1}{s-3} + \frac{1}{(s-3)^2}$$

$$x_1(t) = \mathcal{L}^{-1} \left\{ \frac{1}{(s-3)^2} \right\} = te^{3t}$$

$$x_2(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s-3} + \frac{1}{(s-3)^2} \right\} = e^{3t} + te^{3t} = e^{3t}(1 + t)$$


$$\begin{cases} x_1(t) = te^{3t} \\ x_2(t) = e^{3t}(1 + t) \end{cases}$$

is the solution of the given IVP

Example: Use Laplace transforms to solve the given system of differential equations

$$\begin{cases} x_1'' - 2x_1' - x_2' + 2x_2 = 0 \\ x_1' - 2x_1 + x_2' = -2e^{-t} \end{cases}$$

with the initial conditions

$$x_1(0) = 3, \quad x_1'(0) = 2, \quad x_2(0) = 0$$

Solution:

$$\mathcal{L}\{x_1''\} = s^2 X_1(s) - s x_1(0) - x_1'(0)$$

$$\mathcal{L}\{x_1'\} = s X_1(s) - x_1(0)$$

$$\mathcal{L}\{x_2'\} = s X_2(s) - x_2(0)$$

$$\mathcal{L}\{x_2\} = X_2(s)$$

Replacement in $\begin{cases} x_1'' - 2x_1' - x_2' + 2x_2 = 0 \\ x_1' - 2x_1 + x_2' = -2e^{-t} \end{cases}$

$$[s^2 X_1(s) - 3s - 2] - 2[s X_1(s) - 3] - [s X_2(s) - 0] + 2X_2(s) = 0$$

$$[s X_1(s) - 3] - 2X_1(s) + [s X_2(s) - 0] = -\frac{2}{s+1}$$

$$\begin{cases} (s^2 - 2s)X_1(s) - (s - 2)X_2(s) = 3s - 4 \\ (s - 2)X_1(s) + sX_2(s) = \frac{3s + 1}{s + 1} \end{cases}$$

$$\times s \quad \begin{cases} (s^2 - 2s)X_1(s) - (s - 2)X_2(s) = 3s - 4 \\ (s - 2)X_1(s) + sX_2(s) = \frac{3s + 1}{s + 1} \end{cases}$$

$$-X_2(s^2 + s - 2) = \frac{-2s - 4}{s + 1} \longrightarrow X_2(s + 2)(s - 1) = \frac{2(s + 2)}{s + 1}$$

$$X_2(s) = \frac{2}{(s - 1)(s + 1)} \quad \text{for } s \neq -2$$

$$\text{or } X_2(s) = \frac{2}{s^2 - 1}$$

$$\diamond \text{ You also can use Cramer's Rule} \longrightarrow X_1(s) = \frac{3s^2 - 4s - 1}{(s - 2)(s^2 - 1)}$$

Decompose
the fractions

$$\frac{3s^2 - 4s - 1}{(s - 2)(s - 1)(s + 1)} = \frac{A}{s - 2} + \frac{B}{s - 1} + \frac{C}{s + 1}$$

$$\frac{2}{(s - 1)(s + 1)} = \frac{A}{s - 1} + \frac{B}{s + 1}$$

$$X_1(s) = \frac{1}{s - 2} + \frac{1}{s - 1} + \frac{1}{s + 1} \longrightarrow x_1(t) = e^{2t} + e^t + e^{-t}$$

$$X_2(s) = \frac{1}{s - 1} - \frac{1}{s + 1} \longrightarrow x_2(t) = e^t - e^{-t}$$


$$\begin{cases} x_1(t) = e^t + e^{2t} + e^{-t} \\ x_2(t) = e^t - e^{-t} \end{cases}$$

In-Class Exercises

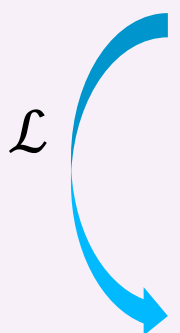
Exercise 1: Use the Laplace transform to solve the initial-value problem

$$y' + y = e^{-3t} \cos 2t, \quad y(0) = 0$$

Recall

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

Solution:


$$y' + y = e^{-3t} \cos 2t,$$
$$\mathcal{L} \left\{ y' + y \right\} = \mathcal{L} \left\{ e^{-3t} \cos 2t \right\}$$
$$sY(s) + Y(s) = \frac{s + 3}{(s + 3)^2 + 4} \longrightarrow Y(s)(s + 1) = \frac{s + 3}{(s + 3)^2 + 4}$$
$$\longrightarrow Y(s) = \frac{s + 3}{(s + 1) [(s + 3)^2 + 4]}$$

Decompose it

$$\frac{A}{s + 1} + \frac{B(s + 3) + C}{(s + 3)^2 + 4}$$

$$Y(s) = \frac{1/4}{s+1} + \frac{-\frac{1}{4}(s+3) + \frac{1}{2}}{(s+3)^2 + 4}$$

$$Y(s) = \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{4} \cdot \frac{s+3}{(s+3)^2 + 4} + \frac{1}{2} \cdot \frac{1}{(s+3)^2 + 4} \quad \xrightarrow{\mathcal{L}^{-1}} \quad ?$$

$$\downarrow$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s - (-1)}\right\} = e^{-t}$$

$$\downarrow$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 2^2}\right\} = \cos 2t$$

$$\downarrow$$

$$\mathcal{L}^{-1}\left\{\frac{2}{s^2 + 2^2}\right\} = \sin 2t$$

Finally,

$$y(t) = \frac{1}{4}e^{-t} + \frac{1}{4}e^{-3t} (\sin 2t - \cos 2t)$$

The solution of the given IVP

Exercise 2: Use the Laplace transform to solve the initial-value problem

$$y'' + 2y = 4t, \quad y(0) = 0, \quad y'(0) = 0$$

Recall

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

Solution:

$$\begin{array}{l} y'' + 2y = 4t, \\ \mathcal{L} \curvearrowright s^2 Y(s) + 2Y(s) = \frac{4}{s^2} \implies Y(s) = \frac{4}{s^2(s^2 + 2)} \xrightarrow{\text{Decompose it}} \frac{As + B}{s^2} + \frac{Cs + D}{s^2 + 2} \end{array}$$

$$Y(s) = \frac{2}{s^2} - \frac{2}{s^2 + 2}$$

$$y(t) = 2t - \sqrt{2} \sin(\sqrt{2} t)$$

$$\mathcal{L}^{-1} \curvearrowleft$$

Exercise 3: Use the Laplace transform to solve the initial-value problem

$$y'' + 4y' + 6y = 1 + e^{-t}, \quad y(0) = 0, \quad y'(0) = 0.$$

Recall

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

Solution:

$$\mathcal{L} \curvearrowright y'' + 4y' + 6y = 1 + e^{-t},$$

$$s^2 Y(s) - \cancel{sy(0)} - \cancel{y'(0)} + 4[sY(s) - \cancel{y(0)}] + 6Y(s) = \frac{1}{s} + \frac{1}{s+1}$$

Simplify it

$$(s^2 + 4s + 6)Y(s) = \frac{2s + 1}{s(s+1)}$$

$$\Rightarrow Y(s) = \frac{2s + 1}{s(s+1)(s^2 + 4s + 6)}$$

Decompose it

$$\frac{A}{s} + \frac{B}{s+1} + \frac{C(s+2) + D}{(s+2)^2 + 2}$$

$$Y(s) = \frac{1/6}{s} + \frac{1/3}{s+1} + \frac{-\frac{1}{2}(s+2) - \frac{2}{3}}{(s+2)^2 + 2}$$

$\mathcal{L}^{-1}$

$$y(t) = \frac{1}{6} \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} - \frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+2)^2 + 2} \right\} - \frac{2}{3\sqrt{2}} \mathcal{L}^{-1} \left\{ \frac{\sqrt{2}}{(s+2)^2 + 2} \right\}$$

$$= \frac{1}{6} + \frac{1}{3} e^{-t} - \frac{1}{2} e^{-2t} \cos \sqrt{2} t - \frac{\sqrt{2}}{3} e^{-2t} \sin \sqrt{2} t.$$

More Exercises ...

Exercise 1: Find the solution to the following initial value problems

$$(1) \ x'' + 9x = 6e^{3t}, \ x(0) = x'(0) = 0;$$

$$(2) \ x^{(4)} + x = 2e^t, \ x(0) = x'(0) = x''(0) = x'''(0) = 1.$$

Exercise 2: Find the solution to the following initial value problems

$$(1) \begin{cases} x_1' + x_2' = 0, \\ x_1' - x_2' = 1, \end{cases} \quad x_1(0) = 1, x_2(0) = 0;$$

$$(2) \begin{cases} x_1'' + 3x_1' + 2x_1 + x_2' + x_2 = 0, \\ x_1' + 2x_1 + x_2' - x_2 = 0, \end{cases} \quad x_1(0) = 1, x_1'(0) = -1, x_2(0) = 0;$$

Formulas for Laplace Transform of Derivatives

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}\{y'''(t)\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$$

$$\mathcal{L}\{y^{(4)}(t)\} = s^4Y(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)$$

Table of Laplace Transforms

	$f(t)$	$\mathcal{L}(f)$		$f(t)$	$\mathcal{L}(f)$
1	1	$1/s$	7	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
2	t	$1/s^2$	8	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
3	t^2	$2!/s^3$	9	$\cosh at$	$\frac{s}{s^2 - a^2}$
4	t^n ($n = 0, 1, \dots$)	$\frac{n!}{s^{n+1}}$	10	$\sinh at$	$\frac{a}{s^2 - a^2}$
5	t^a (a positive)	$\frac{\Gamma(a + 1)}{s^{a+1}}$	11	$e^{at} \cos \omega t$	$\frac{s - a}{(s - a)^2 + \omega^2}$
6	e^{at}	$\frac{1}{s - a}$	12	$e^{at} \sin \omega t$	$\frac{\omega}{(s - a)^2 + \omega^2}$